

Physical Attacks on a UAV System: Overview and Emerging Methods

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Abstract—With the widespread adoption of UAV technology, the physical attacks targeting UAVs have become increasingly diverse, garnering growing attention. Physical attacks pose significant threats to the security of critical hardware within UAV systems, potentially leading to severe consequences such as crashes or unauthorized hijacking. Therefore, conducting in-depth research into physical attack methods on UAV systems not only provides theoretical support and strategic guidance for designing defense measures but also facilitates the optimization and tool-based application of existing attack techniques, paving new pathways for the development of anti-UAV technologies. This review begins with a systematic decomposition and detailed introduction of UAV systems from the perspective of hardware functional structures. Subsequently, it delves into vulnerabilities of UAV systems when facing physical attacks and provides a comprehensive review of existing physical attack methods. Particular attention is given to evaluating the effectiveness, technical characteristics, strengths, and limitations of these methods. Additionally, the review explores emerging physical attack techniques and the potential security threats posed by hardware extensions of UAVs in novel application domains. Furthermore, this review proposes a quantitative risk assessment framework for UAV security, systematically evaluating various physical attack methods based on attack cost, effectiveness, and likelihood. Finally, the review discusses future research directions in the domain of physical attacks on UAV systems, emphasizing the need to enhance existing technologies to strengthen anti-UAV capabilities and highlighting the importance of developing comprehensive defense strategies against physical attacks.

Index Terms—Physical attacks, security, UAV systems, emerging physical attack methods, security risk assessment.

I. INTRODUCTION

THE application prospects of UAVs are increasingly extensive, and their significance is continually highlighted. In aerial photography and film production, UAVs offer unique perspectives for movies, advertisements, and tourism promotion [1]. In agriculture, UAVs equipped with various sensors

enable precision farming by monitoring crop growth, fertilization, and irrigation, thereby enhancing agricultural productivity [2]. In emergency response and disaster monitoring, UAVs can rapidly and flexibly search and survey affected areas, providing timely rescue and monitoring services [3]. Additionally, UAVs demonstrate significant potential in scientific research [4], environmental protection [5], and logistics and transportation [6].

The efficient operation of UAVs relies on their intricate hardware and software systems [7]. The hardware comprises components such as the flight control system (FCS), motors and propellers, sensors, and remote controllers, all working together to accomplish flight missions. The FCS acts as the “brain”, sensing the environment through sensors such as gyroscopes, accelerometers, and magnetometers, and adjusting the flight posture accordingly. Motors and propellers generate lift and thrust. The remote controller communicates control commands wirelessly, while the communication system manages data transmission, and the obstacle avoidance system utilizes LiDAR to detect obstacles. On the software side, flight control algorithms, navigation systems, and image processing enable autonomous flight, automatic obstacle avoidance, and advanced navigation capabilities. Sensors and actuators work closely together, with sensors providing real-time data and actuators adjusting flight status based on this information, ensuring stable and safe performance of UAVs across various application scenarios.

Despite the extensive applications of UAV technology across various fields, there are growing concerns about UAV physical attacks [8], [9], [10], [11], [12], [13]. **Physical attacks** on UAVs primarily target their physical components, including sensors, actuators, and electronic systems. These attacks aim to disrupt, deceive, steal, or manipulate the UAV by damaging its hardware through direct contact or non-contact physical methods. Contact physical attacks refer to adversarial actions involving direct physical access to the UAV body, where critical components are tampered with or damaged, resulting in abnormal system behavior. Non-contact physical attacks involve the use of electromagnetic, acoustic, or optical means to remotely interfere with the UAV’s sensing or control systems without physical contact, thereby disrupting its normal operation. In contrast, **cyber attacks** on UAVs do not directly damage the hardware but instead focus on the system’s wireless communication links and software. By interfering with, deceiving, or stealing communication data, cyber attacks can disrupt the communication between the UAV and the ground control station. Additionally, by injecting malicious code into the UAV’s software, attackers

Received 1 September 2024; revised 17 January 2025, 14 June 2025, and 20 October 2025; accepted 27 November 2025. Date of publication 19 December 2025; date of current version 10 March 2026. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant ZYTS25068; in part by the National Natural Science Foundation of China under Grant 62272366, Grant 62125205, and Grant 62232013; and in part by the Open Project of State Key Laboratory for Novel Software Technology under Grant KFKT2025B77. The Associate Editor for this article was A. Jolfaei. (*Corresponding author: Xiaomin Wei.*)

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Digital Object Identifier 10.1109/TITS.2025.3641708

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can manipulate its operating system or flight control programs. Physical attacks can occur through methods designed to obstruct, hinder, or disable the target UAV. Such attacks pose direct threats to UAVs, impacting public safety, privacy, and data security. This paper primarily considers the following types of **non-contact physical attacks** on UAV systems:

- **Acoustic, Optical and Radio interference:** This involves using physical methods, such as radio jammers, acoustic disruptors, or optical interference, to interfere with the UAV sensors and electronic systems.
- **Electromagnetic Interference:** This involves using electromagnetic interference (EMI) devices or High Energy Electromagnetic Pulses (HEMP) to disrupt or damage the UAV sensors and electronic systems.

With the rapid advancement of UAV technology, methods of physical attacks on UAVs are becoming increasingly diverse and evolving. This trend has garnered significant attention in the academic community, where researchers are conducting in-depth analyses of known UAV physical attack methods and striving to predict emerging new types of attacks. Scholars are focused on exploring novel attack techniques, including those targeting critical UAV hardware components such as FCSs, controller circuit boards, sensors, communication systems, and motors. The exploration of new technologies for UAV physical attacks spans multiple fields, including electronic engineering, computer science, and materials science.

The review of physical attack methods and emerging techniques on UAV systems holds significant importance in the domain of UAV security. This paper focuses on a single UAV system and aims to elucidate the necessity of such a review from two critical perspectives: the application of physical attack techniques for **anti-UAV operations**, and the comprehensive understanding of attack mechanisms to enhance UAV system security and resilience **against physical attacks**. Anti-UAV refers to a systematic defense mechanism that employs a combination of technological means to detect, identify, track, disrupt, capture, or neutralize UAVs that are illegal, hostile, or pose potential security threats. The ultimate goal is to safeguard airspace and ensure operational security.

Reviewing physical attack methods provides valuable insights for developing effective **anti-UAV technologies**. By understanding the principles and effects of existing attack techniques such as jamming and spoofing, security professionals can formulate effective strategies to neutralize malicious UAVs. For instance, knowledge of how to physically interfere with UAVs can guide the design of defense systems to protect sensitive areas and critical infrastructure from unauthorized UAV intrusions. Moreover, this review can inform the development of legal and regulatory frameworks for anti-UAV technologies, ensuring their deployment is both effective and compliant with safety standards.

This review allows for a deep understanding of the principles and implementation steps of various attack techniques, which is crucial for identifying potential vulnerabilities and weaknesses in UAV systems. It not only aids in designing more robust and secure UAVs but also inspires the development of new protective technologies and strategies **against physical**

attacks. For example, improving the materials and structural design of UAVs can enhance their resistance to interference; upgrading sensors and control systems can strengthen their ability to respond to environmental changes and physical attacks. The review can also drive technological innovation by proposing more effective countermeasures, such as intelligent defense systems and automated response mechanisms, thereby significantly enhancing the security and resilience of UAV systems against physical attacks.

Many reviews focus exclusively on cyber attacks against UAVs or Robotic vehicles (RVs) [14], [15], [16], [17], [18], [19], [20], [21], [22]; some reviews address partial aspects of physical attacks [23], [24], [25], [26], [27], [28], [29], [30]; and others focus on physical layer attacks related to UAV communications [31], [32], [33], but there is no dedicated review covering all types of physical attacks on UAV systems. Since UAVs and RVs share significant structural and functional similarities, many physical attack methods applicable to RVs can also be used against UAVs. Therefore, this review will include some physical attack methods used against RVs in its analysis of UAV physical attack methods.

This review addresses the following key issues:

- What is the structure of UAV systems?
- What vulnerabilities exist in a UAV system?
- What types of physical attacks are UAV systems susceptible to? How effective are their attacks, and what are the advantages and disadvantages?
- What are the emerging physical attack methods for UAV systems? What are the threats they pose?
- How to comprehensively evaluate the risk of physical attacks on UAV systems based on multiple factors?
- What are future research directions of physical attacks on UAV systems?

The organizational structure of the article is as follows. The UAV system description and security objectives are provided in Section II. We analyze vulnerabilities of a UAV system in Section III. Next, we overview existing physical attacks on UAVs and discuss their effectiveness, advantages and disadvantages in Section IV. Then, we summarize emerging physical attacks that may occur in the novel application and discuss their threats in Section V. We propose a risk assessment framework and evaluate physical attacks in Section VI. In Section VII, we discuss future research directions from the perspectives of anti-UAV measures and enhancing UAV security. At last, Section VIII concludes this review.

II. UAV SYSTEM DESCRIPTION AND SECURITY OBJECTIVES

A. UAV System Description

UAVs are complex systems composed of various hardware components that work in coordination to enable the completion of diverse tasks. The structure of a UAV system is shown in Figure 1. The hardware components of a UAV system are categorized into eight classes based on functions and explained in the following:

1) **FCS:** The FCS is the key component of the UAV, responsible for managing and controlling its flight, including

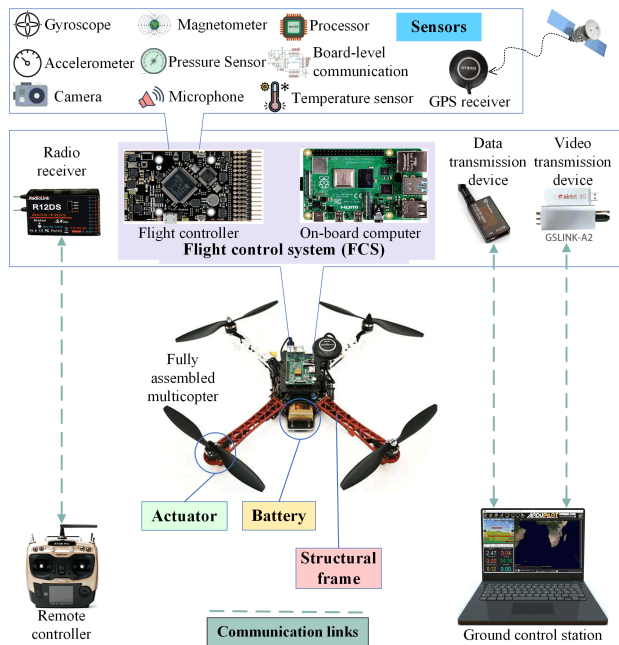


Fig. 1. The structure of a UAV System.

navigation, attitude control, and automatic stabilization. This system encompasses the flight controller, on-board computer, sensor I/O interfaces, and flight control software.

Circuit board-level communication in UAV controllers involves the transfer of data and commands between various UAV components. The flight controller exchanges real-time data with sensors, propulsion systems, communication modules, and other subsystems via an internal bus system. This communication ensures the stability and responsiveness of the UAV, supporting real-time monitoring and control, including attitude adjustments, position tracking, and task execution. Additionally, this internal communication mechanism optimizes data processing and system integration, thereby enhancing the overall performance and reliability of the UAV.

2) *Common Sensors:* Sensors play a critical role in UAVs by perceiving the surrounding environment and providing real-time environmental data for navigation, attitude control, and obstacle avoidance. UAVs are equipped with various sensors, including gyroscopes, accelerometers, magnetometers, barometers, GPS, cameras, and LiDAR.

- **Inertial sensor:** Inertial sensors are a class of devices used to measure an object's motion state and orientation, primarily including gyroscopes and accelerometers. Usually, a gyroscope and an accelerometer are combined as an Inertial Measurement Unit (IMU). The IMU typically utilizes Micro-Electromechanical Systems (MEMS) technology. Both the gyroscope and accelerometer employ a similar mass-spring structure. When the UAV is in motion, the sensing mass vibrates continuously, altering the capacitance and generating electrical signals. These signals are then digitized by an analog-to-digital converter (ADC), resulting in the output of linear and angular rates.
- **Gyroscope:** The UAV gyroscope is a critical IMU. It contains small rotating masses or piezoelectric crystals that can sense changes in angular velocity along the rotational

axes of the UAV, and it is affected by external rotations. By measuring these changes and converting them into electrical signals, the gyroscope provides accurate attitude information of the UAV along three axes. Its primary function is to monitor the rotational state of the UAV in real-time, delivering essential attitude and dynamic information to the navigation system.

- **Accelerometer:** The accelerometer is a critical sensor embedded within the UAV. The accelerometer consists of tiny mass blocks or piezoelectric crystals that sense inertial forces resulting from changes in acceleration and convert these forces into electrical signals. However, these measurements can be affected by external vibrations and shocks. By measuring these electrical signals, the accelerometer accurately detects acceleration along the three axes of the UAV. Its primary function is to provide real-time dynamic data to the navigation system, enabling the UAV to adjust its speed and direction instantaneously.
- **Camera:** The camera is a sensor specifically designed for aerial photography and monitoring, featuring a high-resolution imaging sensor, optical lens, and image processing unit. This camera captures images or video to provide users with clear, detailed aerial views. Its functionalities extend beyond recording landscapes and terrain to include applications in mapping, agricultural monitoring, environmental observation, and infrastructure inspection. Some UAV cameras are equipped with advanced features such as zoom, autofocus, and multi-spectral imaging to meet various task requirements. By delivering high-quality visual data, UAV cameras play a crucial role in numerous applications across different fields. Cameras are instrumental in both civilian and military applications, enabling the capture of images and video recordings. Additionally, they assist pilots in navigating indoor missions by providing critical visual information.
- **Magnetic sensor:** The magnetometer is a crucial sensor for UAVs, designed to detect the Earth's magnetic field, and is typically based on MEMS technology. It consists of tiny magnetic elements that sense changes in the geomagnetic field during UAV rotation and convert these changes into electrical signals. By measuring these signals, the magnetometer provides accurate orientation information relative to the Earth. Its primary function is to deliver real-time magnetic field data to the navigation system, assisting in maintaining stable flight attitudes. The magnetometer usually works in conjunction with other sensors, such as gyroscopes and accelerometers, through sensor fusion techniques to provide comprehensive flight status awareness, which is crucial for navigation, attitude control, and flight stability.
- **Optical flow sensor:** The optical flow sensor for UAVs is a specialized sensor designed for aerial vehicles, consisting of a camera and a processing unit. The camera captures images in real-time by observing textures or feature points on the ground. The processing unit analyzes the displacement of feature points between successive image frames to measure the UAV's motion speed and

direction relative to the ground. The primary function of the optical flow sensor is to provide real-time ground motion information, aiding in tasks such as hovering, positioning, and obstacle avoidance. It plays a crucial role in scenarios where GPS signals are unavailable or during low-altitude hovering, offering precise aerial positioning and flight stability for the UAV.

- **Millimeter wave radar:** The onboard millimeter-wave radar is a crucial sensor mounted on the UAV, consisting of an antenna, transceiver module, and signal processing unit. Operating in the millimeter-wave frequency band (typically between 30 GHz and 300 GHz), this radar provides high-resolution target detection due to its short wavelength. Its primary function is to achieve high-precision detection and distance measurement of the surrounding environment by emitting millimeter-wave signals and receiving the reflected signals from targets. The onboard millimeter-wave radar is widely used in UAV applications for terrain mapping, obstacle detection, target tracking, and autonomous navigation in complex environments. Its high precision, real-time capability, and reliability in various weather conditions make it an essential component for UAV perception and navigation.
- **LiDAR:** The onboard LiDAR is an advanced sensor installed on the UAV, consisting of a laser emitter, receiver, rotating lens, and data processing unit. This radar achieves high-precision three-dimensional mapping of the surrounding environment by emitting laser beams and measuring the time it takes for the reflected laser to return. Its primary function is to provide detailed data on terrain, buildings, and object contours for navigation, mapping, and environmental perception. Onboard LiDAR is widely used in fields such as agriculture, urban planning, and infrastructure inspection, with its high precision and resolution making it a crucial tool for UAV-based sensing and surveying tasks.
- **GNSS receiver:** GNSS (Global Navigation Satellite System) is an overarching term that refers to a network of satellites providing positioning and timing information globally [34], [35], [36]. GPS is one of the most widely used GNSS systems, developed and maintained by the United States. While GPS is a specific instance of a GNSS, GNSS encompasses other systems as well, such as the Russian GLONASS, the European Galileo, and the Chinese BeiDou.

Most UAVs utilize the GNSS receiver for outdoor applications to obtain geospatial References from satellites within their coverage area. The UAV GNSS receiver is a critical navigation device, consisting of an antenna that receives GNSS signals from celestial satellites and a module that processes these signals. Its primary function is positioning and timing, achieved by receiving signals from the GNSS satellites to calculate the UAV's precise location, speed, and altitude. This information is crucial for the UAV's navigation, trajectory planning, and hovering tasks. Typically, the UAV GNSS receiver works in conjunction with other sensors, such as gyroscopes, accelerometers, and magnetometers, using sensor fusion

techniques to provide comprehensive flight state awareness, thereby supporting accurate navigation and flight in various application scenarios.

- **Ultrasonic sensor:** The UAV ultrasonic sensor is a crucial distance-measuring device that typically consists of the ultrasonic emitter and receiver. This sensor operates by emitting ultrasonic pulses and measuring the time it takes for pulses to reflect back from the target object to calculate the distance. The primary function of the UAV ultrasonic sensor is to provide distance perception of the surrounding environment, which is essential for obstacle avoidance, hovering, and precise positioning. This sensor is particularly suitable for close-range operations, offering real-time detection and avoidance of obstacles. Typically integrated into the navigation system, ultrasonic sensors enhance the safety of the UAV during low-altitude flights.

3) *Emerging Sensors:* UAVs are equipped with a diverse array of sensors essential for executing their flight missions. These sensors are integral to the UAV system's functionality, measuring various physical parameters of the environment, such as altitude, speed, and GPS location. The data collected by these sensors is transmitted directly to the flight controller, which uses this information to determine the appropriate actions or maneuvers. In the last subsection, we have discussed common sensors used in UAVs, noting that each application type is supported by a specific set of onboard sensors.

As the demand for UAV applications increases, new types of sensors such as pressure sensors, microphones, temperature sensors, CCD image sensors and LCD touch screens are expected to be integrated into UAVs, significantly enhancing their sensing capabilities and adaptability.

- **Pressure sensor:** In UAV applications, pressure sensors, including barometers, are vital for measuring altitude, monitoring atmospheric conditions, and estimating angles and airspeed [37]. They provide real-time data on the UAV's elevation relative to its surroundings by detecting changes in atmospheric pressure. This information is essential for altitude control, stabilization, and navigation, allowing the UAV to maintain a consistent flight altitude, perform precise altitude-based maneuvers, and adapt to varying environmental conditions. Pressure sensors also play a key role in autonomous landing procedures and environmental monitoring.
- **Microphone:** A microphone is a sensor designed to convert sound signals into electrical signals. Its structure typically includes a diaphragm, a coil, and a magnet. When sound waves reach the microphone, the diaphragm vibrates in response. This vibration affects the nearby coil, causing it to generate an induced current within the magnetic field. The changes in this induced current are converted into a voltage signal, which becomes the electrical output of the microphone. In UAV applications, microphones play a crucial role in enabling voice communication and audio data collection [38]. They are used for capturing ambient sounds and voice commands, facilitating remote control and interaction with the UAV, and enhancing situational awareness by providing audio feedback from the environment. This capability is essential for

tasks such as autonomous operations, where audio signals can be used to improve the accuracy of object detection and navigation, enable precise UAV landing [38], and for communication in applications like surveillance or search and rescue missions.

- **Temperature sensor:** Temperature sensors in UAVs are crucial for both internal and external applications. Internally, they monitor the temperature of critical components such as batteries, motors, and electronic circuits, ensuring they operate within safe limits to prevent overheating and potential malfunctions. Externally, these sensors can provide data on environmental conditions, helping to adjust flight parameters or trigger safety mechanisms if extreme temperatures are detected. By continuously assessing thermal conditions, temperature sensors enhance the UAV's operational reliability and safety, ensuring optimal performance in various environments.
- **CCD image sensor:** CCD is primarily used for capturing high-quality images and videos, and is the core component of the UAV's camera system. They convert light signals into electronic signals, generating precise image data essential for tasks such as navigation, obstacle avoidance, and aerial photography. CCD sensors provide high-resolution images under various lighting conditions, enhancing UAV performance in applications like mapping, surveillance, and inspection. Their high sensitivity and image quality significantly improve the operational capabilities and effectiveness of UAVs.
- **LCD touch screen:** LCD touch screens in UAVs serve as essential user interfaces, allowing operators to interact with and control the UAV's functions. They provide real-time display of flight data, camera feeds, and navigation maps, enabling precise monitoring and control of the UAV's operations. The touch functionality facilitates intuitive access to settings, flight modes, and other control options, enhancing the ease of use and operational efficiency of the UAV. This integration is critical for tasks that require real-time adjustments and monitoring, such as aerial photography, surveillance, and remote inspections.

4) **Actuators:** Also known as servos, actuators in UAVs typically include motors, servos, and electronic speed controllers (ESCs), which are used to control the UAV's attitude and movement, enabling it to fly.

The PWM (Pulse Width Modulation) actuator for a UAV is a crucial component that consists of a motor and a PWM control unit. The motor serves as the power source, with the PWM control unit adjusting the pulse width of the power signal to regulate the motor's speed and power. This PWM technique allows for precise adjustment of the UAV's flight attitude and speed, enabling flexible flight control. The PWM actuator plays a key role in the UAV's navigation system, providing efficient power management and stable flight performance. Its functions also include enabling hovering, adjusting flight paths, and executing various programmable flight tasks.

5) **Payloads and Accessories:** UAVs can carry various payloads and additional equipment, such as firefighting equipments, spraying equipments, and cargo delivery systems, to perform different tasks.

6) **Battery:** The battery is a critical component that provides power to the motors, FCS, and other electronic devices on the UAV. Common types of batteries used include Lithium Polymer (LiPo) batteries.

7) **Structural Frame (Chassis) and Propellers:** The structural frame of a UAV serves as the skeleton that supports and protects all hardware components, while the propellers are responsible for generating lift and thrust.

8) **Communication System:** The communication system enables real-time control and data transmission between the UAV and the ground control station or the remote controller, typically utilizing radio frequency communication technologies, including wireless communication modules and antennas. The communication link is the target of cyber attacks, so this paper does not consider it.

B. Security Objectives of UAV Systems

The security objectives of UAV systems ensure the reliable and trustworthy operation of critical hardware components under physical threats, encompassing four key aspects.

Integrity aims to ensure that the data and physical states of UAV components—particularly the *FCS*, *sensors*, and *actuators*—are not tampered with, altered, or damaged during operation, transmission, or storage. For example, a GPS spoofing or IMU signal injection attack can compromise the integrity of navigation data, leading to trajectory deviations or instability in flight control. Similarly, actuator signal manipulation may result in unsafe propulsion behavior.

Confidentiality focuses on protecting sensitive data and operational parameters within the UAV system from unauthorized access. This is particularly critical for the *sensor suite* and *FCS*, which handle flight telemetry, mission data, and control commands. A breach in sensor data confidentiality could reveal critical mission details or environmental information, posing both security and privacy risks.

Authentication ensures that all entities within the UAV system—such as *communication links between sensors, actuators, and the FCS*—are verified as legitimate.

Availability guarantees that all UAV subsystems remain functional and responsive when required, even under attack or adverse environmental conditions. This includes ensuring stable power supply from the *battery*, *mechanical robustness of the structural frame and propellers*, and *fault tolerance within the FCS*.

III. VULNERABILITY ANALYSIS OF A UAV SYSTEM

Each hardware component of a UAV may harbor vulnerabilities, potentially jeopardizing the device's security and stability. These vulnerabilities may compromise one or more of the UAV's fundamental security objectives—namely, integrity, confidentiality, authentication, and availability. An analysis of potential vulnerabilities of UAV hardware components is shown in the following.

A. FCS

The FCS, comprising the flight controller, onboard computer, and navigation processor, is highly sensitive to EMI,

radiation exposure, and voltage fluctuations. Such perturbations can corrupt control algorithms or disrupt signal processing, directly compromising the *integrity* and *availability* of the control system.

B. Sensors

Sensors—including GPS, IMU, magnetometers, cameras, and LiDAR—are exposed to multiple physical vulnerabilities such as optical interference, acoustic vibration, electromagnetic radiation, and spoofing. These disturbances can distort measurement accuracy or inject falsified data, leading to navigation and attitude estimation errors. Consequently, sensor vulnerabilities primarily threaten data *integrity* and system *availability*, and in cases of signal hijacking or spoofing, may also jeopardize *authentication*.

C. Actuators

Actuators, including motors and servos, are vulnerable to EMI, frequency interference, and voltage instability. Such disturbances may cause uncommanded movements, loss of thrust symmetry, or total control failure. These issues directly affect *availability* and *integrity*, since actuator output deviations can destabilize the flight control loop.

D. Payloads and Accessories

Payloads—such as cameras and communication modules—are vulnerable to EMI, vibration, and temperature fluctuations, which may cause malfunction or data leakage, compromising *confidentiality* and *integrity*. In addition, unauthorized access to payload links can further threaten *authentication*.

E. Battery System

The UAV battery is susceptible to temperature fluctuations, compression, and vibration, which may cause overheating or short-circuiting. These faults compromise *availability* and *integrity* by disrupting power stability and operational continuity.

F. Structural Frame (Chassis) and Propellers

The airframe and propellers are susceptible to external mechanical stress, collisions, and vibrational fatigue. These vulnerabilities can lead to structural deformation, material fatigue, or failure of propulsion dynamics, which directly affects *availability* by reducing flight stability and control responsiveness.

G. Communication Systems

Communication subsystems (responsible for telemetry, command, and data exchange) are vulnerable to cyber attacks. These threats may cause communication loss, latency, or unauthorized command injection, compromising *availability*, *confidentiality*, and *authentication*.

It is noted that direct-contact physical attacks on the battery, structural frame, and propellers are primarily mechanical in nature and beyond the analytical scope of this study, which focuses on vulnerabilities with systemic implications for UAV control and perception subsystems.

IV. EXISTING PHYSICAL ATTACKS ON UAV SYSTEMS

Currently, there are numerous physical attack methods targeting UAVs, which can affect various UAV components. A summary of these physical attack methods is presented in Table I, with detailed explanations provided in the following sections.

A. FCS Attacks on Board-Level Communication Channel

Recent research has highlighted challenges associated with EMI affecting communication signals on circuit boards. These studies reveal that EMI can directly corrupt the original signals, complicating efforts to implement traditional mitigation techniques that typically focus on separating the signal of interest from the interference [92], [93]. Furthermore, some research has identified that communication between control devices is particularly susceptible to electrical resonance within the communication channels (wires) [94], [95], [96], [97].

As shown in row 1 of Table I, Jang et al. [13] proposed an innovative method that disrupted the communication channel responsible for transmitting IMU sensor data to the UAV's control unit, impairing the control unit's ability to accurately receive sensor data regardless of the sensors' specific attributes. Their experiments revealed that there was a specific frequency at which EMI significantly distorted the sensor data, affecting all sensor types for a given control unit board.

Discussion: EMI targeting the board-level communication in the flight controller can effectively disrupt data transmission between the sensors and the UAV's control unit, leading to significant errors in sensor readings and impaired control functions. The primary advantage of this technique lies in its ability to affect all connected sensors through a single interference channel, making it a potent tool for causing broad disruption. However, its effectiveness is highly dependent on identifying the precise frequencies of interference, and implementing such attacks can be technically challenging.

B. Sensor Attacks

It is important to highlight that the flight controller is incapable of differentiating between legitimate and malicious sensor inputs, despite the advanced design of UAV sensors.

1) *Inertial Sensor Attacks:* Recent studies indicate that both gyroscopes and accelerometers are vulnerable to resonant acoustic interference [8], [9], [10], [11], [98], [99], [100], [101]. Methods of physical attacks on inertial sensors are detailed in rows 2-4 of Table I. Adversaries can generate sound waves with frequencies similar to the spring-mass structure, inducing resonance that forces the sensing mass to move and jamming the sensor's output values.

a) *Gyroscope attacks:* Son et al. [8] demonstrated that by exploiting the fact that a gyroscope's output is susceptible to noise at its resonant frequency, it was possible to disrupt the UAV. They showed that injecting noise at the gyroscope's resonant frequency induced resonance, which degraded the accuracy of the gyroscope and interfered with the UAV's operation.

TABLE I
SUMMARY OF EXISTING PHYSICAL ATTACKS ON UAVS

NO.	Reference	Attacked component	Key technique	Limitation
1	[13]	Board-level communication channel	EMI	Depend on precise interference frequencies
2	[8]	Gyroscope	Using sound waves to induce resonance in MEMS sensors	1) Finding the resonance frequency for each sensor; 2) Adjust the environment for each sensor and carry out separate attacks.
3	[9]	Accelerometer		
4	[10][11]	Gyroscope and accelerometer		
5	[12]	Camera	Resonance of MEMS gyroscope using sound waves	Require precise signal design
6	[39][40][41] [42][43][44]	Camera	Specific sign detection disruption	Need physical alteration of the environment or the use of complex camouflage techniques
7	[41][45][46][47] [48][49][50][51] [52][53][54][55] [56][57]	Camera	Obstacle detection deception	
8	[48][58][59]	Camera	Path marking manipulation	
9	[60]	Camera	Multi-object tracking deception	
10	[61][62]	Magnetic sensor	Using forged magnetic field to interfere with magnetic sensor	
11	[63][64]	Optical flow sensor	Laser pointers and infrared signals	Depend on sensor's sensitivity and environmental conditions, and require specialized device and precise alignment
12	[65]	Millimeter wave radar	Physical patches in the environment	Need precisely designed physical patches
13	[66][67][68]	Millimeter wave radar	Resend the victim's signal, or synchronize with the target radar using the same radar	Need complex equipment and a deep understanding of the target radar configuration
14	[69]	LiDAR	Modify the shape of an object	Require precise design
15	[70]	LiDAR	Adjust real objects to create misleading data points	Require careful setup but may be scenario-specific
16	[71][72][73] [74][75]	LiDAR	Crafting specific pulse waveforms which is the same as that of radar pulse	Complex and equipment-intensive
17	[76][77][78][79] [80][81][82][83] [84][85][86][87]	GNSS receiver	Collect and replay GNSS signals, modify navigation information to forge GNSS signals or adjust satellite pseudo-range to forge GNSS signals	Simple attacks are easy to detect; accurate attacks are more complex and difficult to implement
18	[88]	Ultrasonic sensor	Send fabricated ultrasonic signals	Less precise and unpredictable sensor responses
19	[89]	Ultrasonic sensor	Signal cloaking attack	Demand sophisticated equipment and expertise
20	[90]	Ultrasonic sensor	Blind spot attack	Less controllable and may have inconsistent outcomes
21	[91]	PWM actuator	EMI	Require precise frequency and can be unpredictable

b) Accelerometer attacks: Trippel et al. [9] introduced attacks on accelerometers that involved the output biasing and output control by using acoustic signals at the sensors' resonant frequencies. These methods were subsequently adapted for gyroscopes, leading to the development of side-swing and switching attacks [10], [11].

c) Gyroscope and accelerometer attacks: Wang et al. [10] employed an ultrasonic gun to spoof various inertial sensors, including accelerometers and gyroscopes, at their resonant frequencies, thereby disrupting the connected systems. Similarly, Tu et al. [11] utilized acoustic signals to induce resonance in gyroscopes and accelerometers. They enhanced existing resonance techniques and explored out-of-band signal injection methods to exert adversarial control over embedded inertial sensors.

Discussion: Current technologies for attacking, interfering with, and spoofing MEMS accelerometers and gyroscopes have demonstrated notable effectiveness in compromising their performance. Techniques such as acoustic injection can induce vibrations at specific frequencies, causing erroneous readings in MEMS sensors. EMI can disrupt the electrical signals within these sensors, leading to incorrect data outputs.

Spoofing involves injecting false signals that mimic the expected input, tricking the sensor into reporting fabricated data. These methods have been successfully used in experimental settings to show how sensor reliability can be undermined, which can have serious implications for applications relying on precise motion detection.

The primary advantage of these attack methods is their ability to disrupt the accurate functioning of MEMS sensors without needing physical access to the device. This makes them particularly valuable for scenarios where remote interference is required. However, there are significant disadvantages: these methods often require precise knowledge of the sensor's operating characteristics and can be challenging to implement effectively against well-shielded or robustly designed systems. Additionally, advanced sensors may incorporate countermeasures that detect and mitigate such attacks.

2) *Camera Attack:* The approaches for attacking cameras, as shown in rows 5-9 of Table I, can be categorized into two types:

- **Acoustic adversarial attacks:** Poltergeist method is developed by Ji et al. [12], it introduced acoustic adversarial attacks targeting cameras and computer vision systems by

using specially crafted acoustic signals to disrupt sensor readings. This approach exploited vulnerabilities in the physical design of cameras, leading to misclassifications and misleading interpretations by vision algorithms.

- Visual deception attacks: Camera spoofing attacks can target various objectives and employ different attack strategies [27], which have been applied for RVs. This attack technique can be adapted and applied to UAV systems.
 - Specific sign detection disruption: Many studies focus on disrupting specific sign detection, leading to incorrect classifications and control actions, such as misinterpreting a stop sign as a speed limit sign [39], [40], [41], [42], [43], [44].
 - Obstacle detection deception: Some attacks aim to deceive obstacle detection systems, causing the vehicle to either detect non-existent obstacles [45], [46], [47], [48], [49], [50] or overlook real obstacles [41], [51], [52], [53], [54], [55], [56], [57].
 - Path marking manipulation: Certain attacks manipulate lane markings to alter the vehicle's trajectory [48], [58] [59]. For UAV power patrol applications, they may face such attacks.
 - Multi-object tracking deception: Jia et al. [60] extended sticker-based attacks to multi-object tracking tasks, continuously spoofing obstacle positions.

Discussion: Current camera attack technologies can significantly disrupt vision systems of UAVs and RVs by employing acoustic adversarial attacks (such as Poltergeist) and altering the detection results of traffic signs, obstacle detection systems, and lane markings. These methods can cause vehicles to misinterpret environmental information and produce erroneous control actions. The advantage of these approaches is that they can be implemented remotely without direct contact with the equipment.

However, acoustic adversarial attacks require precise signal design and may be affected by environmental noise; visual spoofing attacks need physical alteration of the environment or the use of complex camouflage techniques, making them difficult to implement. Additionally, these methods often require a deep understanding of the target system's operation, increasing complexity and cost. Despite demonstrating significant disruptive capabilities, practical application still faces technical challenges.

3) *Magnetic Sensor Attack:* The magnetic sensor attack methods are summarized in row 10 of Table I. Attackers can exploit the physical properties of sensors to create malicious samples that interfere with data collection. Shoukry et al. [61] described a non-invasive spoofing attack that targets Antilock Braking Systems (ABS) by manipulating magnetic-based wheel speed sensors (Hall sensors) underneath the vehicle. By positioning a thin electromagnetic actuator near the ABS wheel speed sensors, attackers could generate a disruptive magnetic field that canceled the genuine signal and introduced a counterfeit signal, leading to sensor spoofing and misleading wheel speed measurements. Similarly, Barua and Al Faruque [62] demonstrated a non-invasive attack on Hall sensors in a solar inverter using a magnetic field applied

from a close range, resulting in the shutdown of a weak micro-grid.

Discussion: EMI techniques can disrupt the normal readings of magnetometers by generating strong external magnetic fields, thereby interfering with UAV navigation and positioning functions. Spoofing techniques create counterfeit magnetic signals that deceive magnetometers into perceiving these signals as genuine magnetic field data, leading to incorrect positional output. These techniques enable the remote and covert manipulation of magnetometers. Magnetometer interference techniques are relatively easy to implement [61]; however, spoofing techniques are more challenging and require specialized equipment and precise calibration to execute effectively.

4) *Optical Flow Sensor Attack:* The optical sensor spoofing method is summarized in row 11 of Table I. Davidson et al. [63] proposed a method for optical sensor spoofing that allowed attackers to gain an implicit control channel by using laser pointers to deceive a UAV's optical flow sensor via a physically modified ground plane. Similarly, Park et al. [64] used infrared signals to spoof the optical sensors of an infusion pump, resulting in the delivery of an overdose to patients.

Discussion: Current optical sensor attack techniques, such as using laser pointers and infrared signals, effectively deceive sensors in UAVs, leading to unintended control actions or harmful outcomes. However, these techniques depend on the sensor's sensitivity and environmental conditions, requiring specialized equipment and precise alignment.

5) *Millimeter Wave Radar Attack:* The methods for attacking millimeter-wave radars, as shown in rows 12-13 of Table I, can be categorized into two types:

- Passive attacks: Chen et al. [65] introduced passive attacks on Frequency-Modulated Continuous Wave (FMCW) radars by using physical patches placed in the environment. These attacks are limited by their inability to dynamically change the spoofing location and require custom-designed patches for specific attack goals and environments.
- Active attacks: Ranganathan et al. [66] described early detect/late commit (ED/LC) attacks, which involve listening to and retransmitting a victim's signal to spoof an object's range, but this method does not work against FMCW radars. Miura et al. [67] lowered the cost of radar spoofing by using a replica radar synchronized with the target radar to spoof distances. Sun et al. [68] demonstrated more advanced radar spoofing attacks that could induce dangerous behaviors in RVs, including stalling, hard braking, lane changes, and high-speed crashes through multi-stage and cruise control manipulations. It can also have a bad impact on UAVs.

Discussion: Interfering with onboard millimeter-wave radar systems involves generating strong electromagnetic signals to disrupt radar reception, potentially leading to target detection failures or data inaccuracies. While this method does not require direct access to the radar system, its effectiveness can be unstable and demands precise control over radar frequencies. Conversely, spoofing techniques mislead the radar system by generating counterfeit radar signals, causing it to track false targets. Although this approach is more precise,

it necessitates complex equipment and a deep understanding of the target radar configuration, making it both technically challenging and costly.

6) *LiDAR Attack*: We have identified three main techniques for LiDAR spoofing attacks, as shown in rows 14-16 of Table I.

- **Shape alteration**: Attackers can trick LiDAR systems by designing objects with specific shapes that alter the perceived point cloud data [69]. By manipulating the shapes of these objects, adversaries can distort the LiDAR's environmental interpretation.
- **Object placement**: Another method involves strategically positioning real objects to generate misleading laser points in the LiDAR point cloud. This technique disrupts the accuracy of the point cloud model by creating false data through precise object placement [70].
- **Laser injection**: Attackers use a photodiode to synchronize with the LiDAR, intentionally delaying the receipt of laser pulses. They then introduce fake points into the point cloud by crafting specific pulse waveforms. Previous studies have demonstrated the feasibility of both relaying LiDAR pulses from different locations [71] and controlling fake points within the point cloud [72], [73], [74], [75].

Discussion: Shape alteration modifies object shapes to mislead LiDAR interpretation, requiring precise design but offering limited applicability. Object placement involves positioning real objects to create misleading data points, which requires careful setup but may be scenario-specific. Laser injection introduces fake data points by altering laser pulse waveforms, effective yet complex and equipment-intensive. These methods can disrupt LiDAR systems effectively, but they demand sophisticated equipment and precise implementation.

7) *GNSS Receiver Attack*: GNSS vulnerabilities encompass more than just signal blockage and jamming. Spoofing attacks, which involve generating fraudulent GNSS signals to manipulate the position, velocity, and timing reported by a target receiver, have been successfully executed using inexpensive equipment against a broad range of GNSS receivers.

A GNSS spoofing attack involves the use of counterfeit GNSS signals to replace genuine ones, thereby redirecting a UAV to a false location. Through such attacks, UAVs can be intercepted and controlled by misleading them with fake GNSS data. Various techniques for GNSS spoofing have been developed to target UAVs [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], as shown in row 17 of Table I. In this attack, the perpetrator initially deceives the UAV's GNSS receiver into accepting spoofed signals instead of legitimate ones. Once this transition is made, the attacker controls the GNSS receiver to produce false positional data. This can be achieved through either brute-force methods or more sophisticated techniques involving advanced hardware. Brute-force attacks, however, may result in noticeable disruptions in signal strength or timing, making them more detectable and likely to be discarded.

Spoofing GNSS signals to mislead UAV navigation presents a significant security challenge, especially when targeting multiple UAVs in a swarm while maintaining their formation.

Ceccato et al. [85] proposed a method where spoofing signals were transmitted from the ground using multiple antennas and a multidimensional linear filter, designed to accurately reproduce fake positions across a targeted area.

Guo et al. [86] presented a covert spoofing algorithm for UAVs that integrated GPS and inertial navigation systems, showing that effective spoofing occurs when the fake GPS signal's acceleration matches the difference between the UAV's current and spoofed accelerations. The algorithm avoided detection and crashes by ensuring the spoofed trajectory changes gradually compared to the UAV's reference trajectory, with simulations confirming its effectiveness and the enhanced impact of careful trajectory planning.

Gao and Li [87] proposed a covert GNSS spoofing algorithm designed for UAVs with tightly-coupled GNSS and IMUs, employing a fixed-point four-step strategy to bypass anti-spoofing measures like the least-squares residual receiver autonomous integrity monitoring.

Discussion: GNSS attacks have been utilized in various practical scenarios involving both anti-UAV measures and malicious UAV operations. In anti-UAV strategies, GNSS spoofing or jamming disrupts the navigation of adversarial UAV, causing them to lose control or misnavigate, thereby neutralizing potential threats. Conversely, malicious GNSS attacks target UAVs by misleading their navigation systems, which can redirect or incapacitate them, making them useful for offensive operations in restricted or sensitive areas. The primary advantages of GNSS attacks are their ability to remotely control or neutralize UAVs and their strategic value in both defense and offense. However, these attacks have notable drawbacks, including potential collateral interference with legitimate GNSS users, and legal and ethical concerns. In particular, the use of AI-enhanced GNSS spoofing tools introduces new cyber risks, such as the large-scale automation of attacks and the exploitation of sensitive personal or operational data. As highlighted by Radanliev et al. [102], the deployment of advanced AI resources without adequate oversight may amplify systemic vulnerabilities, necessitating stricter governance frameworks to ensure responsible and secure application in both military and civilian domains. Furthermore, from the perspective of AI ethics, the integration of AI into GNSS attack mechanisms raises critical concerns regarding transparency, fairness, and privacy. As Radanliev [103] emphasizes, the ethical deployment of AI demands clear accountability and safeguards to prevent misuse—especially when such technologies could be repurposed for covert surveillance or autonomous offensive actions. Without embedding ethical principles into AI systems, their use in GNSS spoofing risks undermining public trust and exacerbating cyber vulnerabilities.

Currently, GNSS attack techniques predominantly employ GNSS jamming rather than GNSS spoofing for several reasons. 1) **Complexity and implementation difficulty**: GNSS spoofing requires the precise generation and transmission of counterfeit satellite signals, demanding high technical expertise and specialized equipment, making the attack process complex [76]. In contrast, GNSS jamming is relatively straightforward, involving the transmission of sufficiently powerful signals to interfere with legitimate ones, which is

more direct and easier to implement. 2) Maturity of technology: GNSS jamming technology is relatively mature, with various implementation methods and devices readily available, leading to widespread use. Conversely, while GNSS spoofing technology is evolving, efficient and reliable implementation solutions are still being optimized and explored [79], [85], [86], [87], resulting in less frequent practical application.

8) *Ultrasonic Sensor Attack*: The methods for attacking ultrasonic sensors, as shown in rows 18-20 of Table I, can be categorized as follows:

- Signal injection attacks:
 - Direct fraudulent signals: Attackers send fabricated ultrasonic signals to the sensor, potentially arriving before genuine reflected signals and causing the sensor to detect non-existent objects [88].
 - Replay attacks: Genuine ultrasonic signals are pre-recorded and then replayed intermittently to the sensor, disrupting its distance measurement accuracy.
- Advanced manipulation techniques:
 - Signal cloaking: Attackers intercept the incoming ultrasonic signal, conceal the reflected signal using cloaking techniques, and return a fabricated signal to manipulate the sensor's response [89].
- Exploitation of sensor vulnerabilities:
 - Blind spot exploitation: Thin objects or objects covered with acoustic foam are placed in the sensor's blind spots to evade detection, leading to potential collisions [90].

Discussion: These approaches can lead to false detections or inaccurate distance measurements, impairing the sensor's performance. While these techniques can disrupt sensors without physical contact, they differ in effectiveness and complexity. Fake signal injection and replay attacks are less precise and more variable, whereas advanced cloaking methods offer greater effectiveness but demand sophisticated equipment and expertise. Additionally, exploiting sensor vulnerabilities, such as blind spots, is often less controllable and may not always result in consistent outcomes.

C. Actuator Attacks

The actuator attack method is summarized in row 21 of Table I. Dayanikli et al. [91] demonstrated that EMI could be used to overwrite controller PWM commands, compromising a servo motor and rendering a UAV uncontrollable. Their experiments on a fixed-wing UAV showcased the feasibility of remote manipulation of servomotors. However, the authors acknowledged that this method was not applicable to multi-copters. Jang et al. [13] addressed and overcame most of these limitations.

Discussion: EMI can disrupt UAV actuators by inducing faulty electrical signals, causing erratic behavior and impairing flight performance. While it allows for remote compromise without direct access, EMI attacks require precise frequency targeting and can be unpredictable, making mitigation difficult due to the pervasive nature of electromagnetic fields and the complexity of UAV systems.

TABLE II
SUMMARY OF EMERGING PHYSICAL ATTACKS ON UAVS

NO.	Reference	Attacked component	Key technique
1	[92]	Pressure sensor	EMI
2	[104]	Pressure sensor	Sound wave
3	[105]	Barometer	EMI
4	[106][107][108]	Microphone	EMI
5	[109][110][111] [112][113][114] [115][116][117] [118][119]	Microphone	Malicious voice command injection
6	[120]	Temperature sensor	Rectification attacks
7	[121]	CCD image sensor	Electromagnetic emanation
8	[122]	LCD touch screen	Electromagnetic emanations from touchscreens

V. EMERGING PHYSICAL ATTACK METHODS AND THEIR POTENTIAL THREATS

As the demand for UAV applications increases, new types of sensors such as pressure sensors, microphones, temperature sensors, CCD image sensors, and LCD touch screens are expected to be integrated into UAVs, significantly enhancing their sensing capabilities and adaptability. However, these sensors are also susceptible to physical attacks, as detailed in Table II and the following subsections.

A. Pressure Sensor Attack

The methods of attack on pressure sensors are described in rows 1-3 of Table II. Tu et al. [92] showed a deliberate EMI attack on an inflation pump's pressure sensor, impacting the system's actuation. Barua et al. [104] demonstrated a spoofing attack on pressure sensors located in a negative pressure room using acoustic signals at the sensor's resonant frequency.

Regarding the barometer sensor, Lavau et al. [105] found that reading errors could be induced at different frequencies for both pressure and temperature readings. For instance, in the frequency range of 5.8–5.9 GHz, with a field of around 200 V/m, displayed temperatures could decrease by up to 20 degrees Celsius.

Discussion: Attacking a UAV's pressure sensor can lead to significant operational threats by causing inaccurate altitude readings and disrupting flight stability. Altered pressure data can mislead the UAV's altitude control systems, leading to incorrect altitude adjustments or erratic flight behavior. This can result in unsafe flight conditions, such as difficulties during takeoff or landing, increased risk of crashes, and failure to execute precise maneuvers. Furthermore, disrupted pressure readings can impair the UAV's ability to accurately monitor its environment, affecting tasks like terrain mapping and environmental analysis.

B. Microphone Attack

The microphone spoofing attack aims to covertly manipulate a target system into executing malicious voice commands, escaping detection by normal users, as shown in rows 4-5 of Table II. To achieve this, recent attacks have employed two primary methods:

- EMI attacks: EMI attacks target microphone systems by introducing voltages through electromagnetic waves,

disrupting analog sensors. Kune et al. [106] demonstrated a low-power EMI attack that combined signal injection with an analog sensor, using an audio signal amplified with EMI to interfere with signal processing. and J. L. Esteves Kasmi et al. [107], [108] further showed that EMI could induce sound signals in microphones and affect voice assistant systems, highlighting the vulnerability of analog circuits to such interference.

- **Malicious voice command injection:** This category includes attacks where adversaries create deceptive voice commands that are interpreted by the system as legitimate inputs. Yan et al. [109] utilized inaudible voice techniques [110], [111], [112], [113], employing ultrasonic frequencies to control various features of an autonomous vehicle, such as navigation and entertainment systems. Similarly, Zhou et al. [114] explored audio injection methods [115], [116], [117], [118], [119] to manipulate navigation functions in autonomous driving systems by replaying or injecting malicious audio signals.

Discussion: Compromising a UAV's microphone can severely impact its functionality by interfering with its ability to accurately record and process audio inputs. Such attacks may disrupt communication channels and control mechanisms, allowing adversaries to intercept, manipulate, or obstruct essential voice commands and environmental sounds. This can lead to erroneous audio signal interpretation, reduced situational awareness, and potential control loss over the UAV. Moreover, attackers might exploit these vulnerabilities to inject deceptive audio commands or generate disruptive noise, thereby impairing the UAV's performance in tasks like surveillance and navigation.

C. Temperature Sensor Attack

The temperature attack method is shown in row 6 of Table II. Tu et al. [120] demonstrated that adversaries could exploit temperature-based control systems to produce unforeseen effects without directly interfering with the victim's system or setting off temperature alarms. Their attack utilized an unintended rectification phenomenon within amplifiers to manipulate the direct current (DC) voltage of temperature sensor outputs, thereby controlling the system's responses indirectly.

Discussion: Compromising a UAV's temperature sensors can severely impact its operational safety and reliability. Manipulated or falsified temperature readings might falsely indicate normal operating conditions, leading to overheating of key components such as batteries and motors. This could cause malfunctions, destabilize flight performance, or even result in fires or crashes. Additionally, incorrect temperature data can hinder the UAV's ability to adjust to environmental changes, further jeopardizing its functionality and safety.

D. CCD Image Sensor Attack

As shown in row 7 of Table II, Köhler et al. [121] introduced an innovative signal injection attack targeting CCD image sensors, which were employed in various high-precision fields including professional, scientific, and military applications.

Their research demonstrated how electromagnetic emissions could be exploited to alter the image data captured by CCD sensors, affecting the brightness of individual pixels with high precision.

Discussion: Attacks on a UAV's CCD sensor can severely compromise its performance by affecting its image capture capabilities. This disruption can lead to misinterpretation of the UAV's environment, impairing its navigation and obstacle avoidance functions. Consequently, the UAV may collide with obstacles, deviate from its flight path, or collect erroneous visual information, which undermines its effectiveness in tasks such as surveillance, reconnaissance, and environmental monitoring.

E. LCD Touch Screen Attack

As shown in row 8 of Table II, Jin et al. [122] introduced Periscope, an innovative side-channel attack that leveraged electromagnetic emissions associated with human interaction with touchscreens to deduce sensitive inputs on mobile devices. This approach capitalized on the fact that finger movements across a touchscreen create dynamic electromagnetic coupling, which in turn altered the electromagnetic emissions of the screen. These variations could be detected by remote sensing devices. By correlating these electromagnetic measurements with specific finger movements, Periscope enabled the recovery of input data from the touchscreen.

Discussion: Attacking a UAV's LCD can pose significant threats by disrupting the display of critical information to the operator, potentially leading to misinterpretation of the UAV's status and environment. This can result in operational errors, impaired decision-making, and even loss of control over the UAV. Such attacks can exploit vulnerabilities in the display system to inject false information, create visual distortions, or completely disable the display, thereby compromising the safety and effectiveness of UAV operations.

VI. RISK ASSESSMENT

The primary objective of risk assessment is to quantify the threat level posed by identified physical attacks. Based on security risk assessment frameworks outlined in ISO/IEC 27005:2022 [123] and assessment handbook [124], this study proposes a risk assessment method tailored to physical attacks on UAV systems. For each recognized type of physical attack, the assessment is conducted based on three key dimensions: the cost required to launch the attack, the effectiveness of the attack, and the likelihood of its occurrence. By comprehensively evaluating these factors, the overall risk level associated with each attack can be determined. The specific evaluation criteria for these factors are detailed in Subsection VI-A.

A. Establishment of Attack Cost, Effectiveness and Likelihood Levels

1) **Attack Cost Levels:** This criterion evaluates the resources required to execute an attack, encompassing equipment, technical expertise, and time investment. The cost associated with the consequences of risk events can be used to define different levels of consequence scales [123], while the cost incurred

by an attacker to carry out an attack can similarly be used to establish distinct levels of attack cost scales. The specific definitions are provided as follows.

- Low (1): The attack requires easily accessible equipment with minimal technical expertise and incurs costs below \$100 or equivalent resources.
- Medium (2): The attack necessitates specific equipment and moderate technical skills, with an associated cost ranging from \$ 100 to \$ 1000.
- High (3): The attack demands specialized equipment, advanced technical capabilities, and substantial resource investment, with costs exceeding \$1000.

2) *Attack Effectiveness Levels*: This criterion assesses the potential impact or degree of disruption an attack can cause on the UAV system. This element corresponds to the concept of “risk consequences” in the quantitative risk assessment methodology outlined in ISO/IEC 27005:2022. Drawing on the evaluation framework provided in the standard, this study establishes a classification scheme and scale-based scoring rules for assessing attack effectiveness. Levels and scale values are evaluated as follows:

- Low (1): The attack results in minor disruptions to system functionality, with localized effects that do not compromise flight safety.
- Medium (2): The attack significantly interferes with system functions, potentially impairing navigation, control, or perception capabilities. However, it does not directly lead to loss of control or a crash.
- High (3): The attack causes severe damage to critical systems, potentially resulting in UAV loss of control or a crash.
- Extremely High (4): The attack completely incapacitates the UAV system, rendering it uncontrollable or causing catastrophic failure.

3) *Attack Likelihood Levels*: This criterion evaluates the probability of an attack occurring, based on the prevalence of the attack method and the feasibility of execution. Drawing upon the definition of “likelihood” in the ISO/IEC 27005:2022 standard, this study establishes a classification scheme and quantitative scoring criteria for evaluating the likelihood of physical attacks against UAV systems. The specific details are as follows.

- Rare (1): The attack requires exceptionally high technical expertise, and the conditions for execution are highly restrictive, with a likelihood of less than 1%.
- Unlikely (2): The attack demands moderate technical expertise and resources but is constrained by specific conditions, with a likelihood between 1% and 10%.
- Possible (3): The attack method is well-known, and the necessary equipment or tools are widely available, with a likelihood between 10% and 50%.
- Likely (4): The attack is straightforward to execute, involving widely available and inexpensive equipment, with a likelihood between 50% and 90%.
- Almost Certain (5): The attack is highly feasible, using well-established methods that are easy to deploy, with a likelihood exceeding 90%.

TABLE III
RISK ASSESSMENT RESULTS

Attack Type	C	E	L	Risk Score
GNSS receiver attack	1.5	3	4	30
Inertial sensor attack	2	4	3	24
Camera attack	1.5	2	4	20
Magnetic sensor attack	2	3	3	18
Optical flow sensor attack	2	2.5	3	15
Ultrasonic sensor attack	2	2.5	3	15
LCD touch screen attack	1	1	5	15
CCD image sensor attack	2	2	3	12
Microphone attack	1	1	3	9
Board-level communication attack	3	4	2	8
LiDAR attack	3	2.5	3	7.5
Millimeter-wave radar attack	3	3	2	6
Pressure sensor attack	2.5	1.5	2	4.5
Temperature sensor attack	2	2	1	4
Actuator attack	3	4	1	4

C: Cost, E: Effectiveness, L: Likelihood.

B. Attack Cost, Effectiveness and Likelihood Analysis

Metric levels of physical attacks are shown in Table III. The detailed explanation of the level determination is as follows.

1) *Board-Level Communication Attack*: *Cost*: High. It requires EMI equipment and necessitates cracking or physical access to the circuit board, making it highly technically demanding. *Effectiveness*: Extremely High. If the communication system is compromised, it could result in the entire UAV system becoming incapacitated. *Likelihood*: Unlikely. High technical barriers make these attacks unlikely but critical.

2) *Inertial Sensor Attack*: *Cost*: Medium. The attacker must possess specialized knowledge of IMU sensor principles and may require dedicated jamming equipment or directed energy devices. *Effectiveness*: Extremely High. As the IMU is a core component of flight controller, an attack could result in destabilization or crashing. *Likelihood*: Possible. While IMU attacks require specific expertise and resources, their execution is feasible in controlled scenarios or by skilled attackers.

3) *Camera Attack*: *Cost*: Low to Medium. The cost of equipment required for attacking the camera, such as strong light (e.g., laser) or jamming signals, is relatively low. *Effectiveness*: Medium. It can affect the UAV’s visual navigation and target recognition functions, but the impact depends on the importance of the camera. *Likelihood*: Likely. Cameras are commonly exposed and easily targeted, making them more susceptible to attacks in practical scenarios.

4) *Magnetic Sensor Attack*: *Cost*: Medium. It requires specific magnetic interference devices, such as strong magnets or electromagnetic jammers, and necessitates a certain level of expertise. *Effectiveness*: High. Magnetic interference can impair the UAV’s navigation and orientation capabilities, potentially leading to loss of control. *Likelihood*: Possible. Magnetic sensors are sensitive to interference, but attacks require proximity and specific equipment, reducing their frequency.

5) *Optical Flow Sensor Attack*: *Cost*: Medium. It requires knowledge of the optical flow sensor’s working principles and the use of appropriate light sources or interference techniques. *Effectiveness*: Medium to High. It may disrupt the UAV’s stable hovering and positioning functions, though the overall

impact is limited. *Likelihood*: Possible. Attacking optical flow sensors demands precision and detailed knowledge of the UAV's configuration.

6) *Millimeter-Wave Radar Attack*: *Cost*: High. Specialized jamming equipment is needed, making it difficult to execute. *Effectiveness*: High. It may impair the UAV's obstacle avoidance and navigation capabilities, but other methods may be required to achieve significant impact. *Likelihood*: Unlikely. Advanced equipment and technical expertise reduce the practical frequency of this attack.

7) *LiDAR Attack*: *Cost*: High. Professional laser equipment is needed, which can be difficult to obtain. *Effectiveness*: Medium to High. It can interfere with the LiDAR's ranging and navigation functions, but requires precise implementation. *Likelihood*: Possible. Although technically challenging, the increasing use of LiDAR systems may raise its attack surface.

8) *GNSS Receiver Attack*: *Cost*: Low to Medium. Inexpensive GNSS jammers or spoofing devices are readily available on the market and are easy to obtain. *Effectiveness*: High. GNSS attacks can cause the UAV to deviate, lose direction, or even crash. *Likelihood*: Likely. GNSS systems are frequently targeted due to the widespread availability of spoofing and jamming devices.

9) *Ultrasonic Sensor Attack*: *Cost*: Medium. It requires ultrasonic transmitters with specific frequencies, with relatively low equipment and technical barriers. *Effectiveness*: Medium to High. It can disable the UAV's close-range obstacle avoidance capabilities, increasing the risk of collision, especially in low-altitude flights. *Likelihood*: Possible. The increasing use of ultrasonic sensors makes them a viable target for attackers.

10) *Pressure Sensor Attack*: *Cost*: Medium to High. It requires specialized equipment, such as high-frequency noise generators or laser jamming devices, to control or apply external pressure on the environment. *Effectiveness*: Low to Medium. It affects the UAV's altitude control, but the actual impact depends on the intensity of the attack. *Likelihood*: Rare. Pressure sensors are less commonly targeted due to their relatively isolated role in UAV navigation.

11) *Microphone Attack*: *Cost*: Low. Strong sound waves or ultrasonic attacks on the microphone require easily obtainable equipment. *Effectiveness*: Low. It mainly disrupts the UAV's audio sensing functions, with limited overall impact on flight. *Likelihood*: Possible. The ease of obtaining necessary equipment makes microphone attacks plausible in certain scenarios.

12) *Temperature Sensor Attack*: *Cost*: Medium. Requires heating or cooling equipment, with moderate difficulty in execution. *Effectiveness*: Medium. It may affect the UAV's battery management or environmental sensing, but the overall impact is limited. *Likelihood*: Rare. The necessity of precise environmental control reduces the probability of such attacks.

13) *CCD Image Sensor Attack*: *Cost*: Medium. Equipment such as lasers or strong lights is used for the attack, and is easily obtainable. *Effectiveness*: Medium. It may disrupt the UAV's imaging capabilities, interfering with image acquisition and processing. *Likelihood*: Possible. CCD sensors are relatively exposed, increasing the feasibility of attacks.

14) *LCD Touch Screen Attack*: *Cost*: Low. It uses strong light or static interference to disrupt touchscreen operations, with easily obtainable equipment. *Effectiveness*: Low. It primarily affects the ground controller, with limited impact on UAV flight unless the operator is directly affected. *Likelihood*: Possible. Readily available equipment increases the feasibility of such attacks.

15) *Actuator Attack*: *Cost*: High. It requires direct physical contact or in-depth research into the UAV's internal systems. *Effectiveness*: Extremely High. An attack on the actuators can directly cause the UAV to lose control or crash. *Likelihood*: Rare. Actuators are generally well-protected within the UAV structure, reducing their exposure to attacks.

The cost of physical attacks is typically proportional to their effectiveness, but their likelihood also plays a crucial role in assessing overall risk. Low-cost attacks, while often limited in impact, can pose a significant threat if they are highly likely to occur and easy to execute. In contrast, high-cost attacks, despite their technical complexity and potential to cause complete UAV system failure, may be less concerning if their likelihood is low due to the resources and expertise required. Therefore, defense strategies against physical attacks should not only consider the potential resources and capabilities of attackers but also prioritize protection against highly likely, efficient, and easily executable attack methods, which pose a more immediate and probable threat.

C. Risk Assessment

In the context of UAV system security assessment, a comprehensive risk evaluation method is proposed, which quantifies the risk level of different types of attacks. This method calculates an overall risk score based on three key dimensions: Attack Cost (C), Attack Effectiveness (E), and Attack Likelihood (L), as defined in Section VI-A. This risk score allows for informed decision-making regarding security defenses across various attack scenarios. The risk is quantitatively evaluated using the formula 1.

$$RiskScore = (4 - C) \times E \times L \quad (1)$$

The risk score is derived by multiplying the values for each of the three dimensions, resulting in a score ranging from 1 (lowest risk) to 60 (highest risk). This score effectively reflects the potential threat of various attack types to the UAV system.

Based on the computed comprehensive risk scores, as shown in Table III, the threats posed by different attack types can be classified and analyzed. To better understand the varying risk levels, the attacks are categorized into three risk levels:

- **High Risk (Risk Score > 20)**: Attack types in this category pose significant potential threats to the UAV system. If such attacks occur, they could potentially lead to system destabilization or severe damage. For example, inertial sensor attacks and GNSS attacks fall into the high-risk category due to their ability to effectively disrupt UAV navigation and flight stability at moderate or low cost, and with a relatively high probability of occurrence.
- **Medium Risk (Risk Score 15-20)**: These attacks have a moderate threat level and may cause certain disruptions to

the UAV system's performance and functions. Examples include camera attacks, magnetometer attacks, optical flow sensor attacks, ultrasonic sensor attacks, and LCD touch screen attacks, which may cause errors in positioning or impair sensor functions but typically have a more localized effect, with a lower likelihood of occurring.

- **Low Risk (Risk Score < 15):** Attack types in this category have a relatively minor impact on the UAV system, usually affecting auxiliary functions or having a low likelihood of occurrence. For instance, microphone attacks, temperature sensor attacks, and CCD image sensor attacks are considered low-risk, as they primarily affect environmental sensing or ground control equipment, with limited impact on flight safety.

Through the proposed comprehensive risk evaluation model, a clear risk framework for different types of attacks is established. This model not only facilitates the assessment of the potential threat posed by various attacks to UAV systems but also provides a theoretical foundation for devising targeted protective measures. High-risk attacks should be prioritized for defense strategies, while medium and low-risk attacks can generally be mitigated using standard protective techniques.

VII. FUTURE RESEARCH DIRECTIONS

We consider future research directions from following two perspectives: (1) Improving practical methods of physical attacks as anti-UAV techniques (Section VII-A), including EMI attacks and GNSS attacks. (2) Protecting UAVs from physical attacks (Section VII-B).

A. Improving Physical Attacks as Anti-UAV Techniques

1) *Improving EMI Attacks:* EMI has received extensive attention, with related literature indicating that EMI can achieve a variety of attacks, causing interference and damage to various physical components of a UAV system. For example, malicious EMI can launch various sensor attacks [62], [88] [92], [105], [106], [107], [108] [120], [121], [122], [125], actuator attacks [91], and board-level communication attacks [13]. Such interference not only affects the normal operation of devices but also poses significant security risks.

To enhance the practical applicability of EMI in real-world scenarios, it is necessary to develop more precise and efficient EMI attack tools and improve existing countermeasures. This will facilitate the application of EMI technology in anti-UAV and other aviation regulatory fields.

Firstly, current methods often lack precision in the accurate localization and frequency selection of interference signals, resulting in unstable attack effects. To address this shortcoming, higher-precision frequency selection and signal localization techniques can be utilized to improve the accuracy of interference signals, ensuring effective disruption of the target UAV's operations. Secondly, existing EMI attack methods typically interfere with specific frequency bands, overlooking potential threats in other bands. To resolve this issue, AI technology can be integrated to achieve intelligent

multi-band attack capabilities, ensuring simultaneous interference across multiple frequency bands. This will enhance the effectiveness and coverage of the attacks, adapting to diverse UAV communication systems. Through these improvements, the practicality of EMI attack methods and tools will be significantly enhanced, making them more useful for anti-UAV applications in aviation regulation.

2) *Improving GNSS Attacks:* As analyzed in Section VI-C, GNSS attacks pose the highest risk to UAV systems, primarily due to their high effectiveness and high likelihood of occurrence, making them exceptionally threatening in practical applications. However, despite the availability of numerous GNSS attack techniques, research on optimizing and innovating these methods remains relatively limited. Most existing studies focus on implementing attacks using established techniques, while exploration and improvement of novel attack methods are insufficient.

To enhance GNSS attack methods for practical use in aviation regulation and anti-UAV applications [126], [127], several improvements can be made. (1) Precision spoofing: The existing practical GNSS attack methods and tools (not laboratory tools) are mainly based on GNSS interference attack methods. Current practical GNSS attack methods often lack sufficient precision in signal simulation and spoofing, leading to unstable attack effects. It is necessary to enhance the precision of signal simulation technology by using more advanced algorithms and equipment to deceive UAV navigation systems without raising suspicion. This involves creating high-fidelity replicas of GNSS signals that can be selectively targeted at specific UAVs. (2) Portability: Present GNSS attack equipment is often bulky, making it inconvenient to carry and deploy. There is a need to design portable and lightweight equipment to ensure easy carrying and rapid deployment, enabling usage in various environments. (3) Multi-navigation system attack capability: Current methods typically target only a single GNSS system, such as GPS, while neglecting other navigation systems (e.g., GLONASS and Galileo). We should enhance the capability to attack multiple navigation systems simultaneously, ensuring the ability to disrupt or spoof various systems, thereby increasing the effectiveness of the attack. By refining these aspects, GNSS spoofing can become a more viable and effective tool for aviation authorities to manage and mitigate unauthorized UAV operations, enhancing airspace security and safety.

B. Designing Countermeasures Against Physical Attacks

To enhance the physical attack resilience of UAV systems, the following countermeasures can improve the physical attack defense capability of UAV systems.

1) *Attack Shielding:* UAV sensors are often inadequately shielded from environmental factors and malicious interference, which can compromise their functionality. It is necessary to enhance shielding and filtering of sensors to protect them from EMI, environmental extremes, and other forms of tampering [8], [11] [62], [92] [128]. We can employ advanced shielding materials and design techniques to improve resilience.

2) *Attack Detection*: UAVs often lack effective mechanisms to detect and respond to physical tampering or unauthorized access attempts. Advanced tamper-detection systems and real-time monitoring should be integrated to identify and respond to physical tampering, such as, sensor redundancy methods [89], sensor fusion techniques [90], control invariant-based monitors [129], [130], [131], control parameters [132], random variation verification mechanism for sensor parameters [133], and machine learning-based detection [134], [135]. Utilizing sensor data, flight control data, and alert systems to provide real-time warnings of potential security vulnerabilities to UAV systems and their operators. Simultaneously, integrating automated response mechanisms to promptly address threats and effectively mitigate potential damage.

3) *Attack Recovery*: Many UAV systems lack robust incident response and recovery protocols, which can delay or hinder recovery after an attack. We should develop and implement comprehensive incident response plans and automated recovery mechanisms [136], [137], [138], [139], [140], [141], [142], [143]. Ensure that UAVs have the capability to quickly detect, assess, and recover from attacks, including fail-safes and redundancy to maintain operational continuity.

4) *Security Objective-Based Countermeasures*: To mitigate the effects of physical attacks on UAV hardware, defense strategies can be formulated based on the system's core security objectives.

a) *Integrity*: The system should protect FCS commands, sensor data, and actuator signals from unauthorized modification to maintain accurate control and prevent unsafe maneuvers.

b) *Confidentiality*: All critical data and control signals must be encrypted to prevent unauthorized access and safeguard operational information.

c) *Authentication*: Robust authentication mechanisms should verify users, devices, and data sources to ensure that only legitimate entities can access or control UAV components.

d) *Availability*: The UAV system must maintain stable operation of the FCS, sensors, and actuators under environmental stress or EMI to prevent mission disruption.

VIII. CONCLUSION

This review systematically investigates the physical attack techniques targeting UAV systems, aiming to provide theoretical foundations and technical support for the design of defense strategies. By conducting an in-depth analysis of existing physical attack methods, this study not only synthesizes and reinforces the core principles and practical pathways of these techniques but also explores their potential for tool development and practical application, thereby offering more precise and efficient strategies for anti-UAV applications.

The paper begins with a comprehensive introduction to UAV systems, categorizing them based on functional structures and analyzing their key security objectives: integrity, confidentiality, authentication, and availability. Subsequently, it examines the vulnerabilities of UAV systems when faced with physical attacks and systematically reviews existing research on physical attack methods. Particular emphasis is placed on

evaluating the effectiveness, applicability, and strengths and weaknesses of these techniques.

Building upon this foundation, the study expands its scope to include emerging hardware used in UAV applications, exploring novel physical attack techniques and the potential security threats they may pose. To provide a quantitative assessment of the potential impact of various physical attacks, a risk evaluation framework is proposed, incorporating multi-dimensional metrics such as attack cost, attack effectiveness, and attack likelihood. This framework offers a scientific basis for risk management in UAV systems.

Finally, the review concludes by summarizing and envisioning future research directions in UAV physical attack technologies. It emphasizes improving existing attack techniques to enhance anti-UAV capabilities and developing comprehensive defense strategies against physical attacks. These efforts aim to bolster the security performance of UAV systems and promote further advancements in their safety and practicality.

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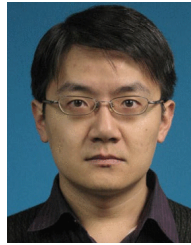
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